Analyzing Transformer Reliability under Domain Shift

Hema Sagar Koppusetti
B.Tech CSE-AI & ML
Lendi Institute of Engineering and Technology
hemasagar3186@gmail.com
github.com/HemaSagarKoppusetti
linkedin.com/in/hema-sagar-koppusetti

***Abstract*—Transformer-based models such as BERT have achieved state-of-the-art performance across many natural language processing tasks. However, their reliability under domain shift remains a critical concern for real-world deployment. This work investigates the robustness and calibration behavior of a BERT sentiment classifier trained on the IMDB movie review dataset and evaluated on both IMDB and SST-2 datasets.**

Our experiments demonstrate that while BERT maintains strong performance under moderate domain shift, prediction confidence and calibration degrade when evaluated on out-of-domain data. Through confidence distribution analysis, reliability diagrams, and error inspection, we provide insights into the behavior of transformer models under distribution shift.

I. Introduction

Transformer architectures have revolutionized natural language processing by enabling large-scale pretraining and contextual language understanding. Models such as BERT have achieved remarkable success on various NLP tasks including sentiment analysis, question answering, and language inference.

Despite their strong performance, a major challenge in deploying such models is their reliability when encountering data distributions different from those seen during training. This phenomenon, known as *domain shift*, can cause performance degradation and unreliable predictions.

In this study, we analyze the robustness of a BERT-based sentiment classifier trained on the IMDB dataset and evaluated on both IMDB and SST-2 datasets. We further analyze confidence calibration and prediction reliability using visualization techniques.

II. Related Work

Transformer-based architectures have become the dominant paradigm in modern natural language processing. BERT [1] introduced bidirectional transformer pretraining using masked language modeling and next sentence prediction objectives, enabling models to learn deep contextual representations of language. This approach significantly improved performance across a wide range of NLP benchmarks.

Following the success of BERT, several studies have investigated the robustness and reliability of deep neural networks under distribution shift. Domain shift occurs when the statistical properties of training data differ from those encountered during deployment. Models trained on one domain may therefore exhibit degraded performance when evaluated on data drawn from a different distribution.

Another important line of research focuses on the calibration of neural networks. Guo et al. [2] demonstrated that modern deep neural networks often produce overconfident predictions, even when their predictions are incorrect. Reliable uncertainty estimation is therefore crucial for building trustworthy machine learning systems.

Recent work has explored techniques such as domain adaptation, calibration methods, and uncertainty estimation to improve the robustness of machine learning models under domain shift. In this work, we focus on empirically analyzing the reliability and calibration behavior of transformer models when evaluated across different sentiment analysis datasets.

III. Methodology

A. Model Architecture

In this study, we employ the **BERT-base-uncased** transformer model for sentiment classification. BERT is based on the transformer encoder architecture and utilizes multi-head self-attention mechanisms to capture long-range dependencies in text.

The model consists of:

- 12 transformer encoder layers
- Hidden representation size of 768
- 12 attention heads
- Approximately 110 million parameters

For classification tasks, the contextual embedding of the special `[CLS]` token is passed through a fully connected layer followed by a softmax function to produce sentiment predictions.

B. Fine-Tuning Procedure

The pretrained BERT model is fine-tuned on the IMDB sentiment classification dataset. During fine-tuning, the parameters of both the transformer encoder and the classification head are updated using labeled training data.

Given an input sentence x, the model predicts the probability of each sentiment class using a softmax classifier:

$$P(y|x) = \text{softmax}(Wh_{CLS} + b) \tag{1}$$

where h_{CLS} represents the contextual embedding of the [CLS] token, and W and b are learnable parameters of the classification layer.

The model is trained using the cross-entropy loss function:

$$L = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{2}$$

where y_i represents the true class label and $\hat{y}_i$ represents the predicted probability.

C. Domain Shift Evaluation

To study model generalization, we evaluate the trained classifier on both the IMDB test dataset and the SST-2 dataset. Since these datasets differ in text length, linguistic style, and sentence structure, this setup provides a natural domain shift scenario for evaluating model robustness.

IV. Experimental Setup

A. Datasets

Two sentiment analysis datasets are used in this study.

IMDB Dataset

The IMDB dataset contains 50,000 movie reviews labeled with positive or negative sentiment. The dataset is balanced and commonly used as a benchmark for sentiment classification tasks. Reviews are typically long-form text containing detailed opinions about movies.

SST-2 Dataset

The Stanford Sentiment Treebank (SST-2) dataset consists of short movie review sentences extracted from larger reviews. Unlike IMDB, SST-2 contains much shorter textual inputs and therefore represents a different linguistic distribution.

The model is trained on the IMDB dataset and evaluated on both datasets to simulate domain shift.

B. Training Configuration

Parameter	Value
Model	BERT-base-uncased
Epochs	3
Batch Size	16
Learning Rate	2×10^{-5}
Optimizer	AdamW
Maximum Sequence Length	128

TABLE I: Training configuration

C. Evaluation Metrics

Model performance is evaluated using several standard classification metrics.

- **Accuracy**: measures overall prediction correctness
- **F1 Score**: balances precision and recall
- **Confusion Matrix**: visualizes prediction errors

In addition, we perform confidence calibration analysis using reliability diagrams and confidence distribution plots to study prediction uncertainty under domain shift.

Dataset	Accuracy	F1 Score
IMDB	0.8826	0.8843
SST-2	0.8589	0.8701

TABLE II: Model performance across domains

V. Results

A. Classification Performance

The results show that BERT maintains strong classification performance even when evaluated on a different dataset. However, a slight performance drop is observed when moving from the training domain to SST-2.

B. Confusion Matrix Visualization

To further analyze classification performance, we visualize the confusion matrices for both the in-domain and domain-shift evaluation datasets.

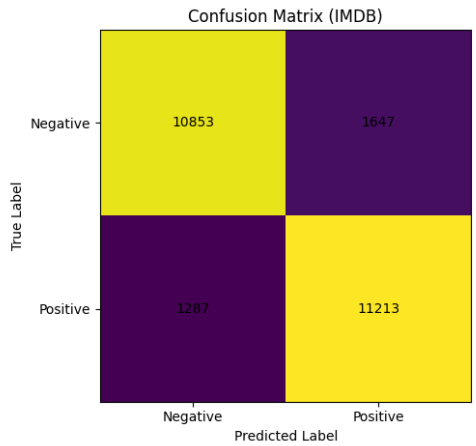

Fig. 1: Confusion matrix for the IMDB test dataset.

The IMDB confusion matrix shows strong classification performance with most samples correctly predicted.

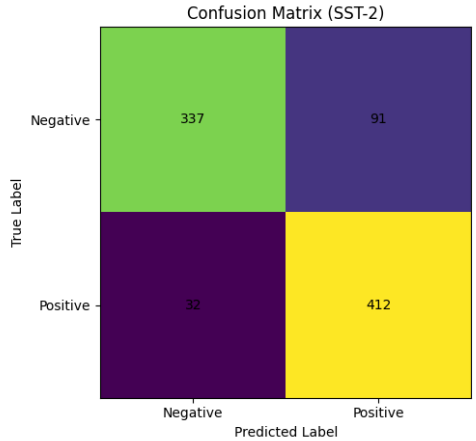

Fig. 2: Confusion matrix for the SST-2 dataset under domain shift.

The SST-2 confusion matrix reveals slightly higher misclassification rates compared to the IMDB dataset, reflecting the effect of domain shift.

VI. Confidence Calibration Analysis

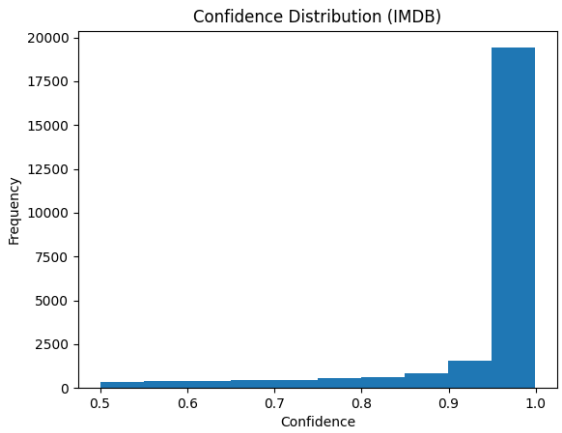

Fig. 3: Confidence distribution for IMDB predictions

The confidence histogram shows that most predictions are associated with high confidence values close to 1.0.

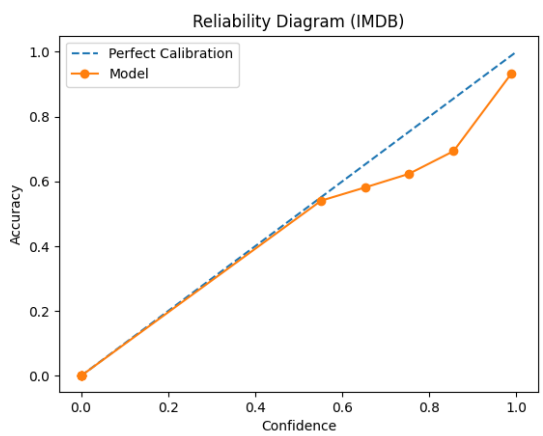

Fig. 4: Reliability diagram for IMDB

The reliability diagram indicates that the model predictions are relatively well calibrated on in-domain data.

Under domain shift, the reliability curve deviates more from the ideal diagonal line, indicating decreased calibration quality.

VII. Error Analysis

Misclassified samples were analyzed to identify common failure patterns. Typical errors occur in the following scenarios:

- Short sentences with limited contextual information
- Sarcastic or ironic expressions
- Mixed sentiment statements

Example:

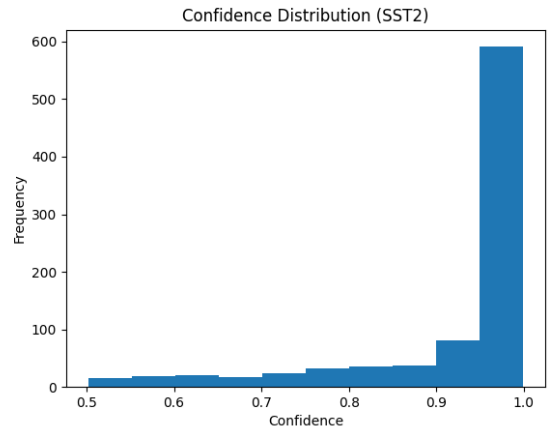

Fig. 5: Confidence distribution for SST-2

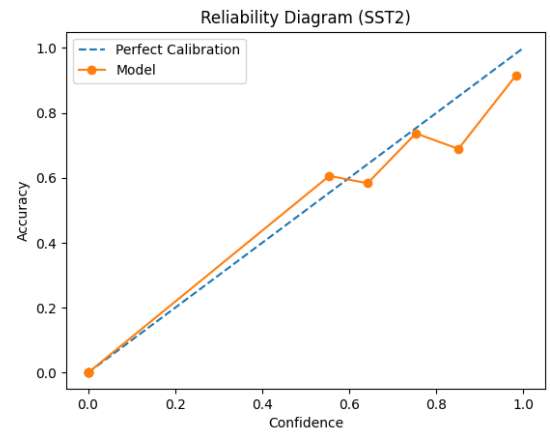

Fig. 6: Reliability diagram for SST-2

”The movie had great acting but a slow plot.”

Such sentences contain both positive and negative signals, making classification more difficult.

VIII. Discussion

The experimental results provide several insights into the behavior of transformer-based models under domain shift.

First, the BERT classifier demonstrates strong generalization capability. Although the model was trained exclusively on the IMDB dataset, it achieves competitive performance on the SST-2 dataset. The observed performance drop between the two datasets is relatively small, indicating that contextual representations learned by pretrained transformer models transfer effectively across related domains.

Second, calibration analysis reveals that prediction confidence behaves differently under domain shift. While the model exhibits relatively good calibration on the IMDB dataset, the reliability diagrams show increased deviation from perfect calibration when evaluated on SST-2. This suggests that although the model continues to make accurate predictions, its

confidence estimates become less reliable outside the training distribution.

Third, the error analysis highlights several linguistic challenges that affect model predictions. Short sentences with limited context, sarcastic expressions, and mixed sentiment statements frequently lead to misclassifications. These patterns suggest that sentiment classification models may benefit from improved contextual reasoning and more diverse training data.

Overall, these findings emphasize the importance of evaluating machine learning systems under realistic deployment conditions. Future research may explore domain adaptation techniques, calibration-aware training methods, and uncertainty estimation approaches to improve the reliability of transformer-based models across domains.

IX. Conclusion

This work investigated the reliability of transformer-based sentiment classifiers under domain shift conditions. A BERT model trained on the IMDB dataset achieved strong performance on both IMDB and SST-2 datasets.

Despite the strong classification performance, calibration analysis revealed increased prediction uncertainty under distribution shift. Future work could explore domain adaptation techniques and calibration methods to improve model reliability.

References

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